

Deep drug-target binding affinity prediction with multiple attention blocks

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Abstract

Drug-target interaction (DTI) prediction has drawn increasing interest due to its substantial position in the drug discovery process. Many studies have introduced computational models to treat DTI prediction as a regression task, which directly predict the binding affinity of drug-target pairs. However, existing studies (i) ignore the essential correlations between atoms when encoding drug compounds and (ii) model the interaction of drug-target pairs simply by concatenation. Based on those observations, in this study, we propose an end-to-end model with multiple attention blocks to predict the binding affinity scores of drug-target pairs. Our proposed model offers the abilities to (i) encode the correlations between atoms by a relation-aware self-attention block and (ii) model the interaction of drug representations and target representations by the multi-head attention block. Experimental results of DTI prediction on two benchmark datasets show our approach outperforms existing methods, which are benefit from the correlation information encoded by the relation-aware self-attention block and the interaction information extracted by the multi-head attention block. Moreover, we conduct the experiments on the effects of max relative position length and find out the best max relative position length value $k \in \{3, 5\}$. Furthermore, we apply our model to predict the binding affinity of Corona Virus Disease 2019 (COVID-19)-related genome sequences and 3137 FDA-approved drugs.

Key words: deep learning; drug-target interaction; self-attention; COVID-19

Introduction

Drugs work by interacting with target proteins to activate or inhibit the biological process of the targets. Thus, identifying novel drug-target interactions (DTIs) is an essential step in the drug discovery field, like drug repurposing [12, 18, 26]. However, transitional costly experiments limit the process to identify new DTIs [26, 28, 31]. Thus, the computational approach for DTI prediction is urgent [37]. Recently, a large of studies proposed

computational methods for DTI prediction. Parts of studies [6, 16, 29, 32] considered the DTI prediction task as a binary classification problem. They focused on the existence of DTI, while some other studies [9, 18, 24] treat it as a regression task to directly predict the binding affinity scores. Here, the binding affinity scores describe the strength of the interactions in drug-target pairs. In this study, we focus on the drug-target binding affinity prediction.

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Recently, deep learning methods are utilized for DTI prediction. DeepDTA [18] proposed a convolutional neural networks (CNNs)-based model for drug representation learning, target representation learning and predicting interaction between them. As one of the widely used deep learning-based models for predicting the binding affinity values, it has achieved an acceptable result. However, it is limited due to that CNN cannot capture the long-distance relationship among atoms in drugs. Based on this, the study [24] introduced a self-attention mechanism-based model with position embedding to encode the relationship among all atoms in compounds. Nevertheless, firstly, it is far from enough to model the compounds since these existing methods just label each atom a corresponding integer according to a dictionary. During modeling the compounds, what is learned is an atom at a specific position. It ignores the correlation between atoms and separates each atom. For example, the compounds 'COC1=C(C=C2C(=C1)CCN=C2C3=CC(=C(C=C3)Cl)Cl)Cl'. In existing methods, given a dictionary {'C' : 1, 'I' : 2, 'O' : 3, etc.}, the character 'C' would be coded as '1' and 'I' is labeled as '2'. The existing methods separated the chloride atom 'Cl' as two fake atoms since they cannot further learn the relative position information between character 'C' and character 'I'. Moreover, the correlation between atoms not only depicts relative position information but also enhances the diversity of atoms. As the 'C' in that example, 'C' atoms would be in any position, but each one includes unique information since their connected atoms are different. Secondly, most existing methods always simply modeled the interaction between drugs and targets by concatenating their representations which is not sufficient to describe the interactions.

Based on these observations, we propose an end-to-end model with multiple attention blocks, named MATT_DTI, to predict the binding affinity scores of drug-target pairs. The protein sequences and SMILES (Simplified Molecular Input Line Entry System) of drugs are the inputs of our proposed model. Firstly, we propose a relation-aware self-attention block to model the drugs from SMILES data, considering the correlation between atoms. The relative self-attention block makes it possible to enhance the relative position information between atoms in compounds while considering the relationship of all elements at the same time. Secondly, two CNN models are utilized to learn the representations of drugs and targets, respectively. Finally, a multi-head attention block is built to model the similarity of drug-target pairs as the interaction information and fully connected networks (FNNs) are used to extract interaction features. Compared with the baseline DeepDTA [18], both of us are sequence representation methods and the protein representation learning part uses the same CNN model. The difference is that we employ a relation-aware self-attention block in drug representation learning to encode correlations of atoms, and a multi-head attention block to model the interaction information of DTIs.

In the experiments, we evaluate our proposed model on two public benchmark datasets, Davis [4] and KIBA [25] datasets, and compare our model with regression-based baselines, KronRLS [19], SimBoost [9], DeepDTA [18] and other recent sequence representation learning methods for DTI prediction. Our MATT_DTI model outperforms these baseline models on Concordance index (CI) and r_m^2 index metrics. Moreover, in order to further investigate the potential of our model, we apply our proposed model to Corona Virus Disease 2019 (COVID-19)-related proteins and list the FDA-approved drugs with high binding affinity scores predicted by our model.

The main contributions of this paper can be summarized as follows.

- (i) In order to model the drug compounds, a relation-aware self-attention block is built to enhance the relative position information between atoms in drugs and capture the long-distance relationship among all the atoms at the same time (section 3 Methods).
- (ii) In order to further extract the interaction information of drug-target pairs, a multi-head attention block is proposed to model the similarity between drugs and target (section 3 Methods).
- (iii) To the best of our knowledge, our results are the state-of-the-art on the two datasets in sequence presentation learning methods for drug-target binding affinity prediction (section 4 Experiments).
- (iv) We apply our model to COVID-19-related proteins and provide a reference to medical experts to find related drugs (section 5 Discussion).

Preliminaries

In this section, we introduce existing approaches for DTI prediction, the background knowledge of attention mechanism and the motivation of this work.

The related studies on DTI prediction

Many studies [6, 16, 29, 32] regarded DTI prediction as a binary classification problem. The proposed models to determine whether the interactions exist between drugs and targets. However, those methods simplified the DTI problem as with chosen binding affinity threshold values [18]. They overlooked the information for the binding affinity value, which describes the strength of the interaction between a drug-target pair. Therefore, the exact way for DTI prediction is directly to predict the binding affinity value based on a regression model.

In recent years, many efforts have been conducted on regression-based models for DTI prediction. The approaches based on random forest algorithm [11, 22] have been successful to predict the binding affinities of drugs and targets. Moreover, similarity-based methods were one option for regression-based DTI prediction, which utilized the similarity information of drugs and targets, such as SimBoost [9] and KronRLS [19].

With the significant success of deep neural networks in the computer version, speech recognition and natural language processing (NLP), many deep learning-based models were proposed to predict DTIs based on regression motivation. In recent works, deep models for DTI prediction mainly include two branches, graph representation method-based approaches with structure information as inputs [15, 30] and sequence representation-based approaches considering sequence information of DTI [18, 34, 35]. In this work, we focus on sequence representation learning approaches DeepDTA [18] and OnionNet [36] proposed CNN-based models for DTI predicting. Especially, DeepDTA focused on the sequence information of both drugs and targets and then used two CNN models for drugs (the SMILES input) and targets (the protein sequence input) as representation learning parts. Then, an information fusion part was connected to predict the binding affinities of drugs and targets. There, three FNN layers were regarded as the information fusion part. Since it could not capture the long-distance relationship between atoms, the study [24] applied a self-attention network (SAN) with position embedding to extract drug representation for DTI prediction.

The background of attention mechanism

Most neural sequence transaction models have an encoder-decoder structure. The transformer [27] is a typical encoder-decoder model based on an attention mechanism. It is widely used in the field of NLP, which has proven the strong ability of transformer in processing text data. The main block of the transformer is the attention function. The attention function can be described as mapping a query (Q) and a set of key-value (K-V) pairs to an output.

Scaled dot-product attention is defined as the generalized attention with Q, K and V. Let the dimension of Q and K be d_k and the dimension of V be d_v

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (1)$$

where $Q \in \mathbb{R}^{n \times d_k}$, $K \in \mathbb{R}^{m \times d_k}$ and $V \in \mathbb{R}^{m \times d_v}$. Attention describes the similarity between the query and each value. The similarity can be measured by inner product of the softmax results and the value. The factor d_k plays a regulatory role so that the inner product is not too large. When Q, K and V are projections from the same inputs, the attention function is the self-attention.

Multi-head Attention is an improved attention mechanism. Firstly, before scaled dot-product attention, the d_{model} -dimensional Q, K and V should be linearly projected h times with learned linear projections to d_k , d_k and d_v , respectively. Specifically,

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad (2)$$

where $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ and $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$. Then, concatenate the results of attention

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_o, \quad (3)$$

where $W_o \in \mathbb{R}^{hd_v \times d_{\text{model}}}$. Here, for each of those Q, K and V, a number of different 'heads' are obtained through linear projections and attention with different weights. Thus, multi-head can be regarded as multiple the same operation in parallel, while parameters are not shared.

Motivation

As seen, when SANs are used on DTI prediction, an atom will conduct an attention operation with all atoms. It leads to SANs disperse the attention distribution to all elements and then overlook the essential correlation between atoms. In addition, most deep models for DTI prediction simply concatenate drug and protein representations to model the interaction between them. The way ignores the interaction features between drug and protein representations. In this study, we propose a deep model on DTI binding affinity prediction, in which the correlation between atoms and the interaction information between drugs and targets are considered.

Methods

In this work, we introduce a multiple attention blocks-based model—MATT_DTI, to predict the binding affinity scores of drug-target pairs, as shown in Figure 1. Like most deep learning-based DTI models, our model consists of three parts: drug representation learning, protein representation learning and interaction

learning. Specifically, we propose a relation-aware self-attention block in the drug representation learning process. The relation-aware self-attention block is to encode correlations by enhancing the relative position information between atoms. Then, two CNN models are utilized to extract features from drugs and proteins in drug representation learning and protein representation learning processes, respectively. Finally, the interaction learning model is exploited to combine and extract interaction features from both drug representations and protein representations by multi-head attention.

Input embedding

The inputs of our model are SMILES sequences for drugs and FASTA sequences for proteins. According to the work [18], the SMILES sequence is comprised of characters representing atoms and structure indicators. Mathematically, a drug is

$$D = \{d_1, d_2, \dots, d_l \dots\}, \quad (1)$$

where $d_i \in N^*$ and the sequence length is varied, which depends on a compound. In this study, we define a hyperparameter l_d to restrict the max input length for drugs. Inspired by the token embedding and position embedding in transformer [27], the input of drug representation learning is the sum of token embedding and position embedding of SMILE sequences. The token embedding $E_t^d \in \mathbb{R}^{l_d \times e_d}$ has a trainable weight $W_t \in \mathbb{R}^{v_d \times e_d}$, where v_d is the vocabulary size of drugs and e_d is the embedding size of drugs. The position embedding $E_p^d \in \mathbb{R}^{l_d \times e_d}$ has a trainable weight $W_p \in \mathbb{R}^{l_d \times e_d}$. The output of the embedding operations is

$$X^d = E_t^d + E_p^d, \quad (2)$$

where $X^d \in \mathbb{R}^{l_d \times e_d}$.

The same as the mathematical expression of drugs, a protein sequence is mathematically expressed as,

$$P = \{p_1, p_2, \dots, p_l \dots\}, \quad (3)$$

where $p_i \in N^*$ and the length of P depends on proteins. We also define a hyperparameter l_p as the fixed protein input length to ensure the same size of inputs. Different from drug sequence, the trainable embedding layer of protein sequence is similar to DeepDTA [18] as

$$X^p = \text{Embedding}(P, e_p), \quad (4)$$

where e_p is the embedding size of protein sequences and $X_p \in \mathbb{R}^{l_p \times e_p}$.

Protein representation learning model

As for protein representation learning process, our proposed model is developed from DeepDTA [18] and the learning model for protein sequences also utilizes three convolutional layers as the feature extractor, followed by a max pooling layer. As for the CNN model in presentation learning model, suppose there exist L_c convolutional layer

$$C^{L_c} = \text{CNN}(X^p), \quad (5)$$

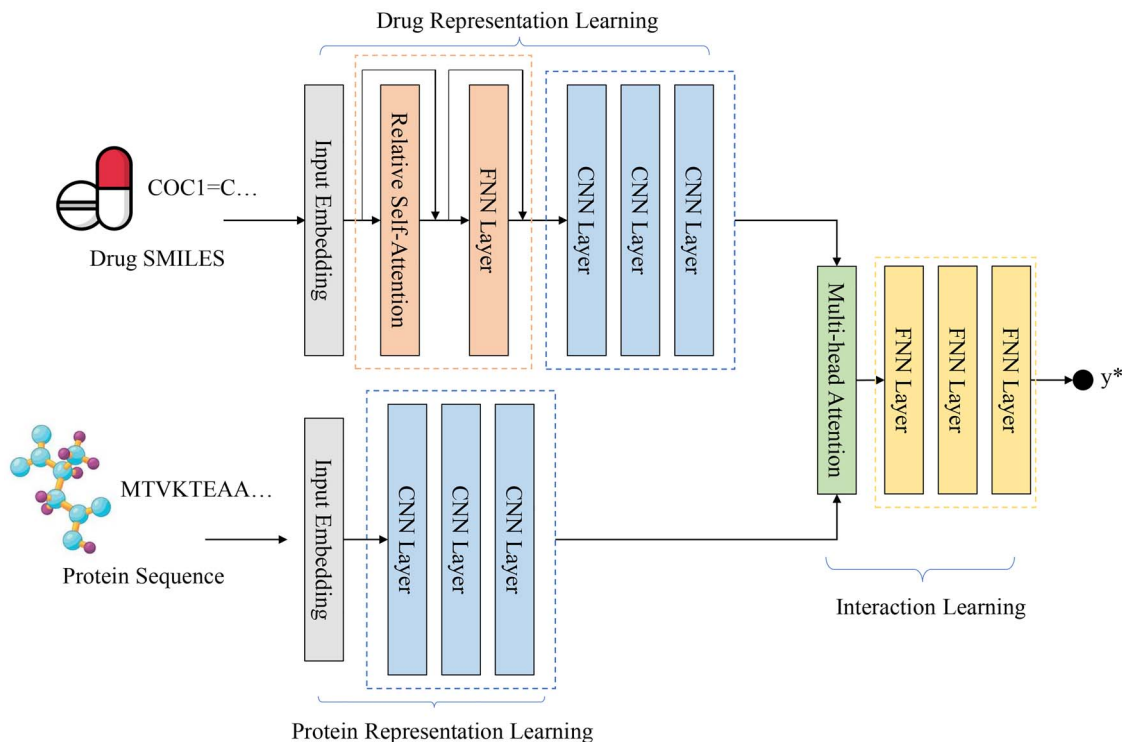


Figure 1. Illustration of our proposed MATT_DTI which considers drug SMILES and protein sequence as input to predict the binding affinity of drug-target pairs, which use a relation-aware self-attention block to strengthen the relative position information when encoding drug compounds and employ multi-head attention to model the interaction of drug representations and protein representations.

where C^{l_c} is the output of L_c th convolutional layer and the l th ($l \in (0, L_c)$) layer can be formally expressed as

$$C_l^l = f\left(\sum_{v=0}^{V-1} C_{l+v}^{l-1} k_v^l\right), \quad (6)$$

where k^l indicates the trainable filters in l th convolutional layer, the size of it is $1 \times V$, C^l is the output of l th layer and $f(\cdot)$ is the activation function. Based on this, when $L_c = 3$ in protein representation learning, the output of protein representation learning could be calculated as

$$R_p^{\text{out}} = \text{Pooling}(\text{CNN}(X^p)), \quad (7)$$

where $\text{Pooling}(\cdot)$ is the max pooling function.

Drug representation learning model

SANs have drawn increasing interest, especially in the NLP field. SANs have the ability to capture long-distance dependencies by explicitly attending to all the elements, regardless of distance [33]. It contributes to representing a drug because SANs capture the long-distance relation between all atoms in a compound. However, SANs have a major limitation that it disperses the attention distribution and thus overlooks relative information of elements [7, 23]. The relative information of atoms in drugs describes the essential correlation between atoms. This leads to that SAN is not sufficient to model SMILES data. Thus, a relation-aware self-attention block is proposed in drug representation learning.

The relation-aware self-attention block

In the field of NLP, self-attention with relative position representations [23] has already considered the pairwise relationship between words. It considers the relative distance information to model the distance between the words during conducting the self-attention. According to work [23], it can be simplified as

$$\text{rel_att_output} = \text{Rel_Att}(Q, K, V, w_r), \quad (8)$$

where $w_r \in N^*$ is the relative distance length between words.

Inspired by this, we developed it to encode the correlation between atoms. We first define k kinds of relative relationships between atoms, which are embedded into learnable parameters $W^R \in \mathbb{R}^{k \times e_d}$. Taken X^d as the input, the output of a self-attention with relative position representations [23] layer can be formally expressed as

$$R_d^{\text{in}} = \text{Rel_Att}(X^d W^Q, X^d W^K, X^d W^V, W^R) \quad (9)$$

$$= \text{softmax}\left(\frac{(X^d W^Q)(X^d W^K)^T + A^R}{\sqrt{e_d}}\right)(X^d W^V), \quad (10)$$

where $W^Q, W^K, W^V \in \mathbb{R}^{e_d \times e_d}$ denote parameter matrices of attention layer. $A^R \in \mathbb{R}^{l_d \times l_d}$ indicates the relationship matrix, of which, A_{ij}^R represents the correlation between the i th and the j th elements

$$A_{ij}^R = (X_i^d W^Q)(W_{\min(j-i, k)}^R)^T. \quad (11)$$

Here, X_i^d is the i th vector in X^d . $\text{clip}(\cdot)$ is employed to select corresponding embedding in W^R . Thus, A^R can be served as inductive biases to revise the attention distribution.

Then, a residual connection [8] and an FNN layer are following. As for the FNN layers in this study, L fully connected layers can be calculated by

$$a^l = \text{FNN}(a^0), \quad (12)$$

where a^0 is the input of FNN layers, the a^l is the output of L th FNN layer and the $l \in (0, L]$ fully connected layer can be described as

$$a^l = f(W_f a^{l-1}), \quad (13)$$

where W_f is the trainable weight and a^l is the output of l th fully connected layer. Therefore, the output of the residual connection and the FNN layer in the relation-aware self-attention block is

$$R_d = \text{FNN}(R_d^{\text{in}} + X_d) + (R_d^{\text{in}} + X_d). \quad (14)$$

Since the protein representation is learned by the CNN model, the three convolutional layers are also used to exploit drug information. We insist the two CNN models could ensure that the drug and protein representations are projected to the same space. Thus, the output is

$$R_d^{\text{out}} = \text{Pooling}(\text{CNN}(R_d)), \quad (15)$$

while $L_c = 3$ in drug representation learning. In this process, layer normalization [2] and dropout [10] are used.

Interaction learning model

The existing way of interaction learning is to concatenate the representations of drugs and proteins. It overlooks the interaction information of drugs and proteins. In similarity-based DTI prediction models [9, 19], the similarity information of drug-protein pairs was used as the interaction information in them. Inspired by this, a multi-head attention block is exploited to model the similarity of drug-protein pairs as the interaction information of them. Here, the drug representations are regarded as the query, while the protein representations are the key and value in the attention mechanism. Mathematically, the output is

$$I_{dp}^{\text{in}} = \text{MultiHead}(R_d^{\text{out}}, R_p^{\text{out}}, R_p^{\text{out}}), \quad (16)$$

with three heads in this study. Then, a residual connection [8] is used as

$$I_{dp} = \text{conc}[g(R_d^{\text{out}}), g(R_p^{\text{out}}), g(I_{dp}^{\text{in}})], \quad (17)$$

where $\text{conc}(\cdot)$ is a concatenation function and $g(\cdot)$ is global average pooling operation. Then, a 3-layered FCN ($L=3$) is employed to learn the interaction information from I_{dp}^{in} and the last layer of the network has only one neuron as the output of our model

$$y^* = \text{FNN}(I_{dp}), \quad (18)$$

Table 1. Summary of the benchmark datasets

	Davis	KIBA
Proteins	442	229
Compounds	68	2111
Interactions	30056	118254
Training data	25046	98545
Test data	5010	19709

where y^* is the predicted binding affinity value of the drug-target pair. The weights of our proposed MATT_DTI model are optimized by the mean square error (MSE) between the network output y^* and the actual binding affinity value y

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i^* - y_i)^2. \quad (19)$$

Experiments

We proposed a novel drug-target binding affinity prediction method based on multiple attention blocks with sequence information of drugs (compounds) and proteins as inputs. In this section, we conducted experiments with our proposed model (MATT_DTI) on two benchmark datasets: Davis [4] and KIBA [25] datasets. The CI and r_m^2 metrics were used to measure the performance of the proposed model and the baseline models.

Benchmark datasets

We evaluated our proposed model on two benchmark datasets, Davis [4] and KIBA [25] datasets. The Davis dataset contains the 442 kinase proteins and their relevant inhibitors (68 ligands) with respective dissociation constant (K_d) value. The K_d values were transformed into log space, as [9, 18], pK_d , as the binding affinity values, which is explained in 20,

$$pK_d = -\log_{10}\left(\frac{K_d}{1e9}\right). \quad (20)$$

The KIBA dataset was developed from the KIBA approach, which comprised 467 proteins, 52498 drugs and their binding affinity scores originally. Here, the KIBA scores measure the kinase inhibitor bioactivities and are regarded as the binding affinity values. SimBoost [9] filtered it to contain 229 unique proteins and 2111 unique drugs for a fair comparison. As for the input of proteins and drugs in the Davis and KIBA dataset, we followed the DeepDTA method [18] in which the SMILES of drugs and protein sequences were digitized to a fixed maximum length by a dictionary. Table 1 summarizes the details of the Davis and KIBA dataset.

Experiments setup

We evaluated the performance of our MATT_DTI on the benchmark datasets. Like the study DeepDTA [18], we firstly clipped the training data as training set and validation set to find the optimal settings of our model, like number of filters, filter length, hidden size, dropout rate and number of epochs. The final results given in this section were the average results on the test set with 5 times training. Table 2 gives the parameter settings in experiments depending on datasets. All models were trained on 1 NVIDIA 2080Ti GPU.

Table 2. Summary of parameter settings for MATT_DTI

parameter	KIBA	Davis
max length (drug)	100	85
max length (protein)	1000	1200
embedding size	128	
number of filters in CNNs	32 64 96	16 32 64
filter size (drug)	8	
filter size (protein)	12	16
hidden size in FNNs	1024 1024 512 1	
batch size	256	128
epoch	300	
dropout	0.1	
optimizer	Adam	
learning rate	0.001	
activation function	ReLU[14]	

Metrics

To evaluate the performance of our model, firstly, the CI was used as the evaluation metrics

$$CI = \frac{1}{Z} \sum_{\delta_i > \delta_j} b(b_i - b_j), \quad (21)$$

where b_i is the prediction value with larger affinity δ_i , b_j is the prediction value for smaller affinity δ_j and Z is a normalization constant. Moreover, the $b(x)$ is the step function [19]

$$b(x) = \begin{cases} 1, & \text{if } x > 0, \\ 0.5, & \text{if } x = 0, \\ 0, & \text{if } x < 0. \end{cases} \quad (22)$$

Then, in order to better evaluate our model, r_m^2 [20, 21], which is widely used in this filed, is the another metric in this work. Mathematically,

$$r_m^2 = r^2 * (1 - \sqrt{r^2 - r_0^2}), \quad (23)$$

where r^2 and r_0^2 are the squared correlation coefficient values between the observed and predicted values with and without intercept, respectively. Only r_m^2 value of a model on test set is larger than 0.5, the model is an acceptable model.

Experiment 1: relation-aware self-attention-based representation learning for drug compounds

Table 3 lists the average results on the drug-target binding affinity prediction tasks. As seen, MATT_DTI improves the prediction quality in both two datasets, reconfirming the necessity of modeling the long-distance relationship and the relative position information of compounds. Besides, our models outperform all the baseline works in all metrics, indicating the superiority of the proposed approaches. In particular, The MATT_DTI with the self-attention block achieves better performance than DeepDTA, revealing the contribution of self-attention that modeling the long-distance relation of all elements in drugs. Moreover, the MATT_DTI with a relation-aware self-attention block (Rel_sa:CNN) outperforms the MATT_DTI model with a self-attention layer (sa:CNN), indicating that modeling the relative

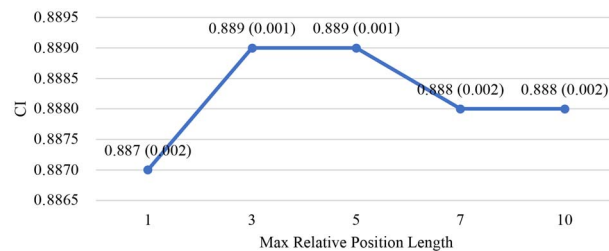


Figure 2. CI results on KIBA dataset. Effects of relative position length in proposed MATT_DTI with a relation-aware self-attention block for drug representation and a multi-head self-attention in interaction learning.

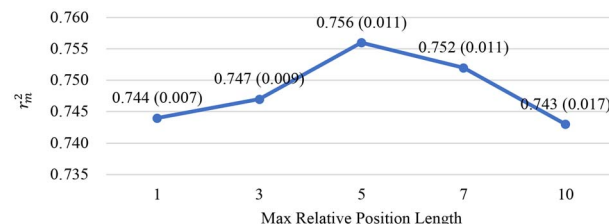


Figure 3. r_m^2 results on KIBA dataset. Effects of relative position length in proposed MATT_DTI with relative self-attention for drug representation and a multi-head self-attention in interaction learning.

information can raise the ability of the self-attention model on capturing the atoms' information.

Experiment 2: interaction learning with multi-head attention

In this section, we conducted the experiment about the interaction learning part based on multi-head attention and compared it with the existing way based on concatenation way. Table 4 gives the average test results on both KIBA and Davis datasets. One intuition of our approach is to capture interaction features via modeling the similarity between drug-protein pairs by a multi-head attention block. To evaluate it, we implemented models with a multi-head attention block in the interaction learning process. As shown in Table 4, the DeepDTA model and MATT_DTI model with a multi-head attention block (MulH_attention + FNN) achieve higher results than the models without it (Concatenation + FNN), revealing that extracting interaction features with multi-head attention is superior to concatenation.

Experiment 3: effects of max relative position length

We finally investigated the effects of relation position length in the relation-aware self-attention block on the drug-target binding affinity prediction task. As shown in Figures 2 and 3, MATT_DTI with max relative position length $k = 5$ has the best performance in the KIBA dataset. As plotted in Figures 4 and 5, we believe the max relative position length with $k = 3$ is superior to other setting for Davis dataset. The different distribution of KIBA and Davis dataset may lead to the slight difference of best max relative position length. As seen, MATT_DTI with max relative position length $k \in \{3, 5\}$ improves the prediction performance, indicating that the correlation between atoms is better modeled with max relative position length $k \in \{3, 5\}$.

Moreover, we compare our proposed model with other sequence representation-based approaches in Table 5. As seen, our final results are higher 0.026 than DeepDTA on KIBA and

Table 3. Test results on KIBA and Davis dataset. The proposed MATT_DTI model includes a relative self-attention block in drug representation learning. The max relative position length k in 'Rel_sa:CNN' is set to 5. In this table, 'sa:CNN' denotes that the model has a self-attention block before CNN layers, while 'Rel_sa:CNN' has a relative self-attention block in front of CNN layers

Dataset	Methods	Compounds	Proteins	Interaction	CI (std)	MSE	r_m^2 (std)
KIBA	KronRLS [19]	Pubchem Sim	S-W	–	0.782 (0.001)	0.411	0.342 (0.001)
	SimBoost [9]	Pubchem Sim	S-W	–	0.836 (0.001)	0.222	0.629 (0.007)
	DeepDTA [18]	CNN	CNN	FNN	0.863 (0.002)	0.194	0.673 (0.009)
	MATT_DTI	sa:CNN	CNN	FNN	0.881 (0.000)	0.162	0.734 (0.002)
	MATT_DTI	Rel_sa:CNN	CNN	FNN	0.889 (0.001)	0.151	0.745 (0.008)
Davis	KronRLS [19]	Pubchem Sim	S-W	–	0.871 (0.001)	0.379	0.407 (0.005)
	SimBoost [9]	Pubchem Sim	S-W	–	0.872 (0.001)	0.282	0.644 (0.006)
	DeepDTA [18]	CNN	CNN	FNN	0.878 (0.004)	0.261	0.630 (0.017)
	MATT_DTI	sa:CNN	CNN	FNN	0.880 (0.003)	0.261	0.632 (0.015)
	MATT_DTI	Rel_sa:CNN	CNN	FNN	0.884 (0.004)	0.254	0.649 (0.009)

Table 4. Test results on KIBA and Davis dataset. The interaction model of the proposed MATT_DTI is based on multi-head attention. The max relative position length k in 'Rel_sa:CNN' is 5

Dataset	Methods	Compounds	Proteins	Interaction	CI (std)	MSE	r_m^2 (std)
KIBA	DeepDTA[18]	CNN	CNN	Concatenation + FNN	0.863 (0.002)	0.194	0.673 (0.009)
	DeepDTA-Attention	CNN	CNN	MulH_Attention + FNN	0.875 (0.003)	0.173	0.724 (0.012)
	MATT_DTI	Rel_sa:CNN	CNN	Concatenation + FNN	0.889 (0.001)	0.151	0.745 (0.008)
	MATT_DTI	Rel_sa:CNN	CNN	MulH_Attention + FNN	0.889 (0.001)	0.150	0.756 (0.011)
Davis	DeepDTA[18]	CNN	CNN	Concatenation + FNN	0.878 (0.004)	0.261	0.630 (0.017)
	DeepDTA-Attention	CNN	CNN	MulH_Attention + FNN	0.877 (0.002)	0.252	0.648 (0.014)
	MATT_DTI	Rel_sa:CNN	CNN	Concatenation + FNN	0.884 (0.004)	0.254	0.649 (0.009)
	MATT_DTI	Rel_sa:CNN	CNN	MulH_Attention + FNN	0.890 (0.003)	0.229	0.682 (0.009)

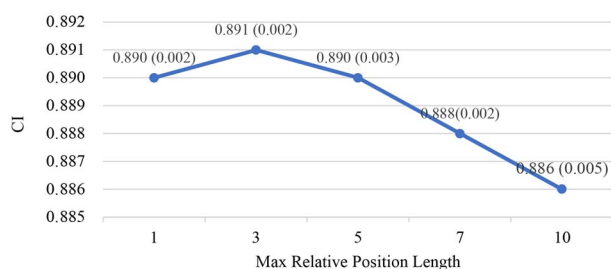


Figure 4. CI results on Davis dataset. Effects of relative position length in proposed MATT_DTI with a relation-aware self-attention block for drug representation and a multi-head self-attention in interaction learning.

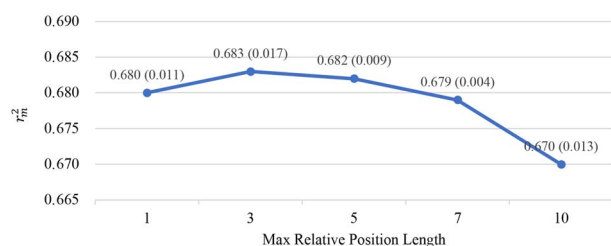


Figure 5. r_m^2 results on Davis dataset. Effects of relative position length in proposed MATT_DTI with a relation-aware self-attention block for drug representation and a multi-head self-attention in interaction learning.

higher 0.012 on Davis for CI metric. The r_m^2 also higher 0.083 than DeepDTA on KIBA and higher 0.052 on Davis. In general, our model performs better than all other sequence representation learning approaches.

Discussion

Recently, the new coronavirus (SARS-CoV-2) infection is spreading rapidly, and the daily incidence rate is increasing worldwide. It is urgent to find out a valid drug for the patient. Drug repurposing is one of the computational efforts that re-utilize FDA approved drugs, or compound succeeded in phase one clinical trials, for a new indication, to take advantage of the proved toxicity [5, 13]. Drug repurposing is regarded as one potential way for finding new coronavirus treatment [3]. Therefore, in this section, we apply our trained model to predict the binding affinity scores between existing drugs and the genome sequences of COVID-19-related severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). We believe that the discussion could provide an example to apply our model in real-life situations and hope our results can provide scientists with an assistant to learn the coronavirus.

Based on studies [1, 3], we extract the genome sequences, 3C-like proteinase, RNA-dependent RNA polymerase, helicase, 3'-to-5' exonuclease, endoRNase and 2'-O-ribose methyltransferase of SARS-CoV-2 from the National Center for Biotechnology Information database; 3137 FDA-approved drugs are included in this section. Table 6 lists parts of the FDA-approval antiviral drugs with top binding affinity values predicted by our MATT_DTI with weights trained by KIBA dataset and existing approaches [1]. The full lists of the 6 genome sequences could be found at supplementary data, see Supplementary Data available online at <http://bib.oxfordjournals.org/>.

Nowadays, the effective drugs to cure COVID-19 have not been found; thus, we cannot verify our results. It is only a theoretical result on drug repurposing task. We just hope that the experiment will reflect the way to use our model in practical applications. Moreover, like studies [1, 3], we hope our work can provide scientists with some ideas for new drugs.

Table 5. Results on KIBA and Davis dataset of our proposed model and the existing baseline methods

Dataset	Methods	CI (std)	MSE	r_m^2 (std)
KIBA	DeepDTA [18]	0.863 (0.002)	0.194	0.673 (0.009)
	MT-DTI [24]	0.882 (0.001)	0.152	0.738 (0.006)
	WideDTA[17]	0.875 (0.001)	0.179	–
	GANsDTA [34]	0.866 (–)	0.224	0.675 (–)
	MATT_DTI	0.889 (0.001)	0.150	0.756 (0.011)
Davis	DeepDTA [18]	0.878 (0.004)	0.261	0.630 (0.017)
	MT-DTI [24]	0.887 (0.003)	0.245	0.665 (0.014)
	WideDTA[17]	0.886 (0.003)	0.262	–
	GANsDTA [34]	0.881 (–)	0.276	0.653 (–)
	MATT_DTI	0.891 (0.002)	0.227	0.683 (0.017)

Table 6. Parts of the FDA-approval antiviral drugs with top affinity scores of 3 genome sequences of SARS-CoV-2 predicted by our model

The genome sequences	Drug	Rank of 3137 drugs
3C-like proteinase	Peramivir	25
	Lopinavir	45
	Saquinavir	54
	Zanamivir	73
	Danoprevir	83
	Ritonavir	89
RNA-dependent RNA polymerase	Saquinavir	58
	Peramivir	120
	Lopinavir	148
	Danoprevir	183
	Daclatasvir (BMS-790052)	202
	MK-5172	243
helicase	Peramivir	51
	Lopinavir	55
	Saquinavir	102
	Elvitegravir (GS-9137)	118
	Danoprevir	137
	Daclatasvir (BMS-790052)	215

Conclusion

In this work, we propose a multiple attention blocks-based model to (i) enhance relative position information between atoms when encoding drugs and (ii) model the interaction between drug representations and target representations. Empirical results of the drug-target binding affinity prediction task on two benchmark datasets demonstrate the effectiveness of our proposed methods. The extensive analyses suggest that (i) encoding the relative position information is beneficial to drug representations, (ii) modeling the interaction can further improve the performance of predicting the binding affinity of DTIs and (iii) the best max relative position length to encode drugs is in 3–5 for the KIBA and Davis dataset. Furthermore, we apply our trained model to predict the binding affinity scores of SARS-CoV-2-related genome sequences and 3137 FDA-approved drugs to provide some reference for COVID-19-related scientists.

- In order to encode the correlation between atoms of drugs, MATT_DTI employs a relation-aware self-attention block to enhance the relative information between atoms when encoding drug compounds.
- In order to extract interaction feature of drug-target pairs, a multi-head attention block is proposed to model the similarity between drugs and target in MATT_DTI.
- Experimental results of DTI prediction on two benchmark datasets show our MATT_DTI outperforms existing models, which is benefit from the correlation and interaction information.
- We further apply our model to FAD-approved drugs and COVID-19-related proteins, which could provide a reference to medical expert.

Key Points

- MATT_DTI is a deep learning-based model for drug-target binding affinity score prediction.

Supplementary data

Supplementary data, including code, weight and results, are available online at https://github.com/ZengYuni/MATT_DTI/.

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